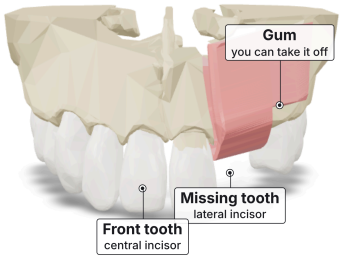
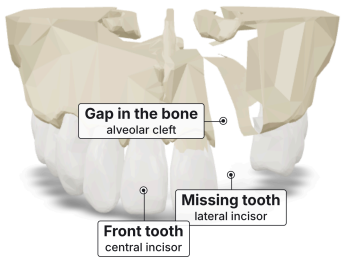
Stage 1: Before surgery

# Bone graft for a cleft in the gum: the three stages

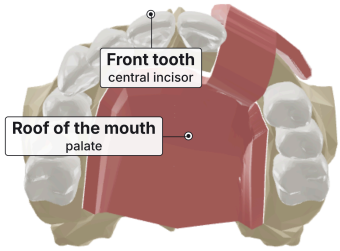
**This model shows the upper jaw of a child born with a cleft on the left side. There is a gap in the bone that holds the teeth. An operation called a bone graft fills this gap with bone.**



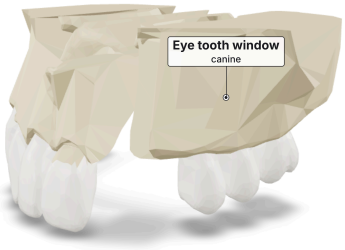

Whole jaw, with the gum




Gum taken off: the gap in the bone




From below: the roof of the mouth




Left side: the eye tooth window

## What this model shows

The gap is in the bone of the gums, next to the front teeth. Its medical name is the **alveolar cleft**.[1]

In this model, one tooth on the left side never grew. It is the small tooth next to the front tooth (the **lateral incisor**). When a tooth never grows, the medical name is **agenesis**. About half of children with a cleft do not have this tooth on the cleft side.[2,3,4]

At about 9 to 10 years old, you have both baby teeth and adult teeth. The adult front teeth and first back teeth are through. The baby eye teeth and baby back teeth are still in place. The adult eye teeth and the teeth behind them are still growing in the bone.[5,6]

The pink part is the gum. It covers the gap. This is how it looks after surgery on the lip and the roof of the mouth in early life.[1] The pictures show the jaw with and without the gum.

## The eye tooth

The adult eye tooth (**canine**) is still high up in the bone next to the gap. It leans toward the gap. Its root is about half grown.[7,8,1] Over the next few years it moves down. It needs bone to grow into.[1] A small window in the bone on the left side lets you see it.

## Why surgeons do the bone graft

The graft aims to:[9,1,10]

- join the two parts of the upper jaw
- give the eye tooth bone to come down into
- hold up the teeth next to the gap and the side of the nose
- close any hole between the mouth and the nose

## When surgeons do the graft

Most teams do the graft at about 8 to 11 years old. They pick the time by how far the teeth have grown, not by age.[10,11,12]

The best time is before the eye tooth comes through the gum. Then its root is about 1/4 to 2/3 grown. Grafting then gives the most bone next to the teeth.[13,1,11]

## The operation

The surgeon takes soft, spongy bone, most often from the hip. They pack it into the gap and close the gum over it.[14,15,10]


**Sources** **1.** Lilja J. Alveolar bone grafting. Indian J Plast Surg. 2009;42:S110-S115. doi.org/10.1055/s-0039-1699383 **2.** Knüppe AL, et al. Developmental characteristics of the permanent upper lateral incisor in unilateral cleft lip and palate. Oral Maxillofac Surg. 2024;28:909-917. doi.org/10.1007/s10006-024-01226-1 **3.** Dentino KM, et al. Is missing maxillary lateral incisor in complete cleft lip and palate a product of genetics or local environment? Angle Orthod. 2012;82:959-963. doi.org/10.2319/012012-49.1 **4.** Suzuki A, Takahama Y. Maxillary lateral incisor of subjects with cleft lip and/or palate: Part 1. Cleft Palate Craniofac J. 1992;29:376-379. doi.org/10.1597/1545-1569(1992)029<0376:MLIOSW>2.3.CO;2 **5.** AlQahtani SJ, Hector MP, Liversidge HM. Brief communication: the London atlas of human tooth development and eruption. Am J Phys Anthropol. 2010;142:481-490. doi.org/10.1002/ajpa.21258 **6.** Hoang L, et al. Factors contributing to canine impaction in unilateral cleft lip and palate patients undergoing alveolar bone grafts. J Oral Maxillofac Surg. 2023;81:1286-1294. doi.org/10.1016/j.joms.2023.06.029 **7.** Kadi H, et al. CBCT assessment of canine eruption and development following alveolar bone grafting in unilateral cleft lip and palate. Cleft Palate Craniofac J. 2023;60:386-394. doi.org/10.1177/10556656211064477 **8.** Caceres Manfio A, et al. Eruption path of permanent maxillary canines after secondary alveolar bone grafting in complete unilateral cleft lip and palate. Am J Orthod Dentofacial Orthop. 2022;161:e416-e428. doi.org/10.1016/j.ajodo.2021.06.019 **9.** Weissler EH, et al. Alveolar bone grafting and cleft lip and palate: a review. Plast Reconstr Surg. 2016;138:1287-1295. doi.org/10.1097/PRS.0000000000002778 **10.** Semb G. Alveolar bone grafting. Front Oral Biol. 2012;16:124-136. doi.org/10.1159/000337666 **11.** Kaura AS, et al. Optimal timing of alveolar cleft bone grafting. J Craniofac Surg. 2018;29:1551-1557. doi.org/10.1097/SCS.0000000000004680 **12.** Fahradyan A, et al. Optimal timing of secondary alveolar bone grafting: a literature review. J Oral Maxillofac Surg. 2019;77:843-849. doi.org/10.1016/j.joms.2018.11.019 **13.** Bergland O, Semb G, Abyholm FE. Elimination of the residual alveolar cleft by secondary bone grafting and subsequent orthodontic treatment. Cleft Palate J. 1986;23:175-205. PMID 3524905. pubmed.ncbi.nlm.nih.gov/3524905 **14.** Boyne PJ, Sands NR. Secondary bone grafting of residual alveolar and palatal clefts. J Oral Surg. 1972;30:87-92. PMID 4550446. pubmed.ncbi.nlm.nih.gov/4550446 **15.** Pan W, et al. Secondary alveolar bone grafting and iliac cancellous bone harvesting. J Craniofac Surg. 2016;27:883-891. doi.org/10.1097/SCS.0000000000002603



This sheet is for a study at Aichi Gakuin University and Nagoya University. Version 0.1, 22 October 2026.